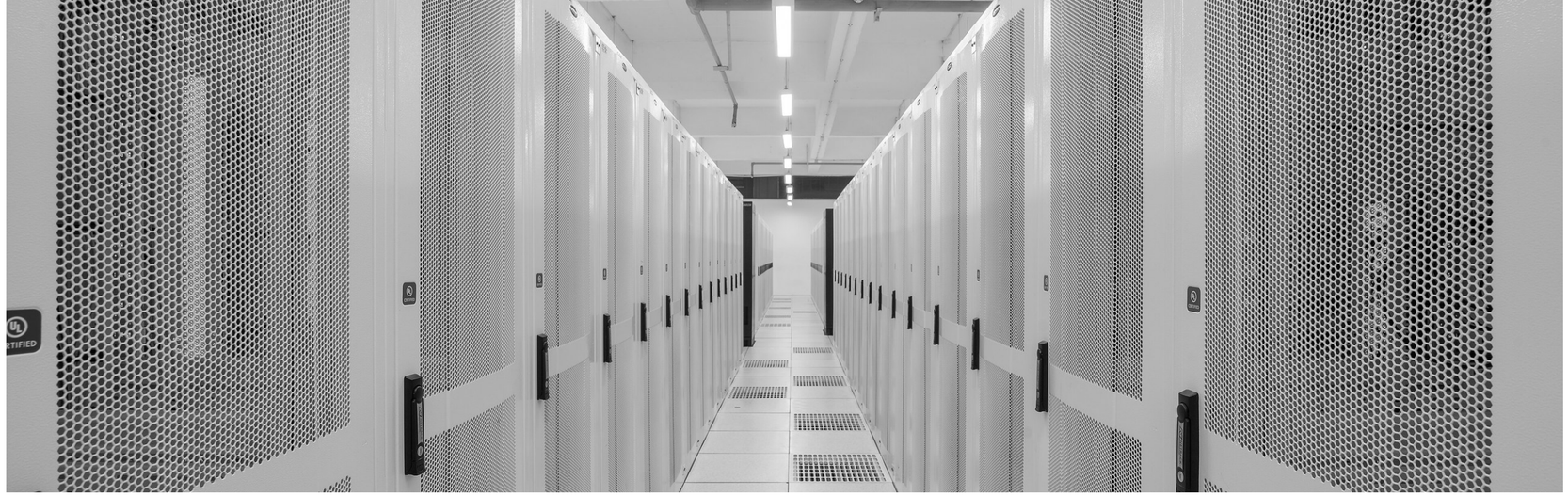


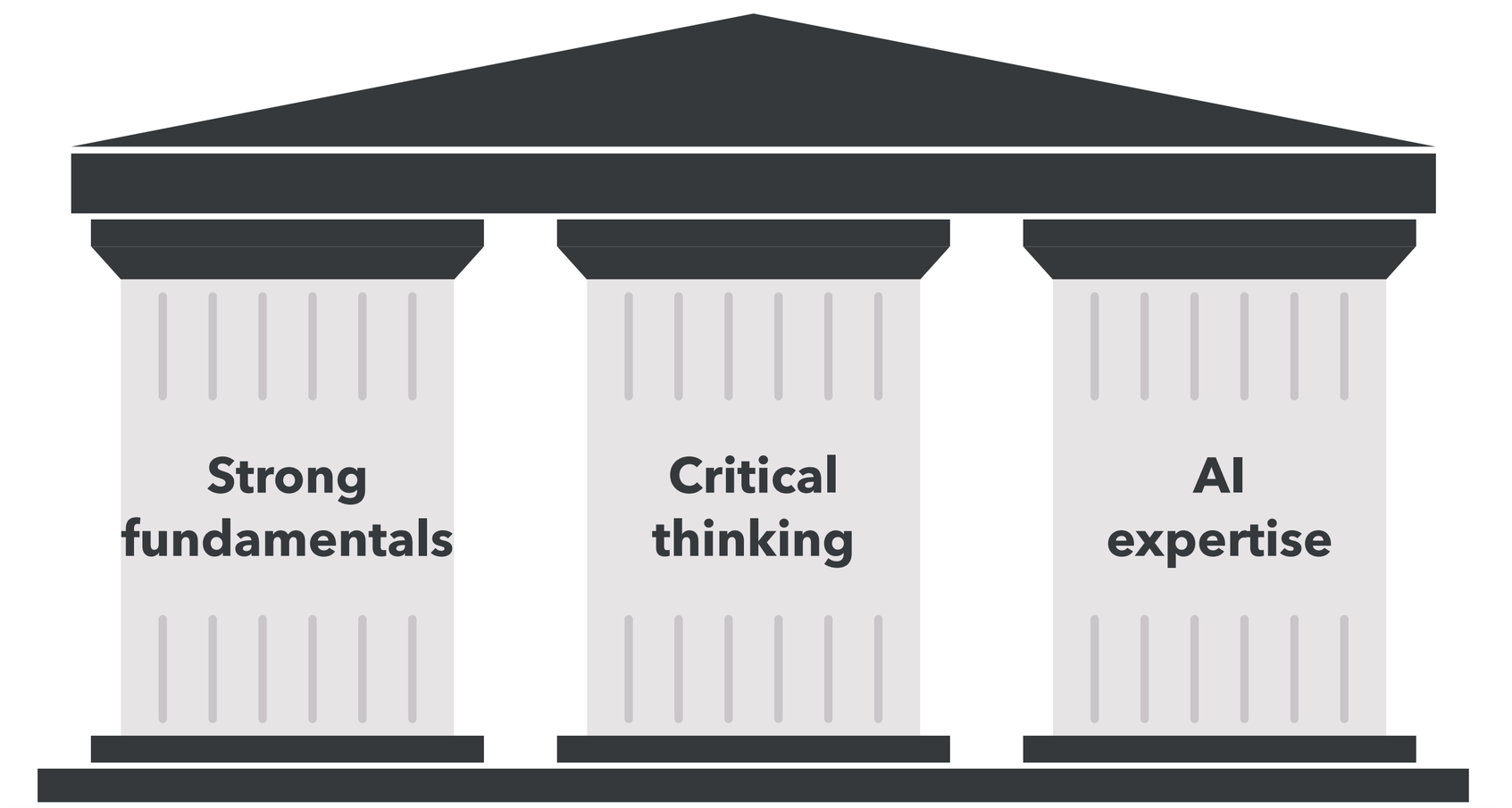
THE FINANCE CURRICULUM IN THE AGE OF AI

SESSION 1

KERRY BACK, RICE UNIVERSITY
DEPARTMENT OF FINANCE, UNIVERSITY OF ZURICH

NOVEMBER 4, 2026

PILLARS OF A FINANCE EDUCATION



OPEN QUESTIONS

1. We can now go faster (and deeper). How much faster?
2. How much Excel?
3. How much Python?

HOW MUCH EXCEL?

- Arrow keys, sum, average, stdev, correl, …
- The value of Excel now is to check AI's answers.
- Example: tangency.kerryback.com

HOW MUCH PYTHON?

- Types, if-else, loops, functions, objects, methods, basic pandas and numpy, what's in what library.
- Little need for Jupyter notebooks.
- Work interactively with AI instead.

AI AND CODING

October 2025

"They just don't work. They don't have enough intelligence … I feel like the industry is making too big of a jump and is trying to pretend like this is amazing, and it's not. It's slop."

January 2026

"I really am mostly programming in English now, a bit sheepishly telling the LLM what code to write… in words."

February–March 2026

"It is hard to communicate how much programming has changed due to AI in the last 2 months … Coding agents basically didn't work before December … I don't think I've typed like a line of code probably since December."

Andrej Karpathy: OpenAI co-founder, former Tesla AI chief, now at Anthropic

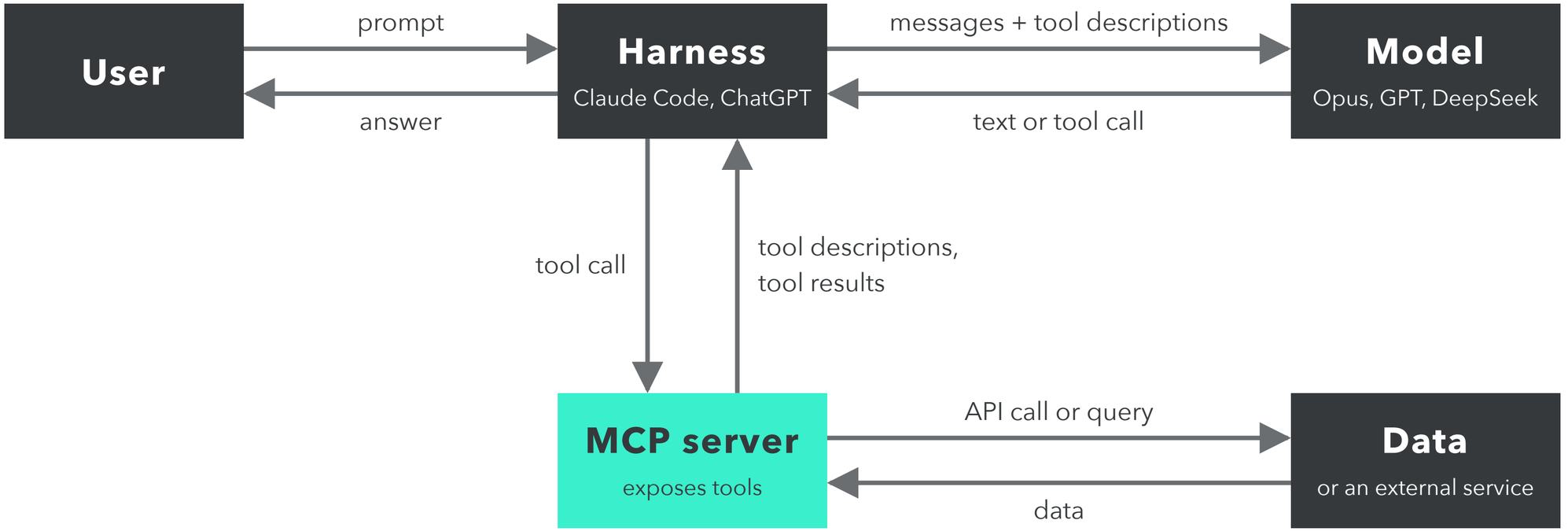
NEW STUDENT DELIVERABLES

- Software apps
- Skills
- Custom agents

MCP servers connect students’ agents to data

DATA

WHAT AN MCP SERVER IS



WHAT “EXPOSES TOOLS” MEANS

- The server gives the harness a text description of each tool: name, purpose, and inputs.
- The harness sends the descriptions to the model with each request.
- The model returns a tool call. The harness sends it to the server, and the server runs the tool.

WHAT A SERVER SENDS FOR ONE TOOL

```
{
  "name": "french_data",
  "description": "One table of a Ken French dataset as CSV. Returns are in decimals.",
  "inputSchema": {
    "type": "object",
    "properties": {
      "name":  {"type": "string"},
      "table": {"type": "integer", "default": 0}
    },
    "required": ["name"]
  }
}
```

> The model sees this text, not the function's code.

Simplified from the server's response to a `tools/list` request.

END-OF-DAY PRICES

Server	Free coverage	Additional paid coverage	Hosted or local
Shibui	US (NYSE, NASDAQ), 64 years; no key; fair-use limits	None listed	Hosted
Equibles	US daily bars; 100 requests/day	10,000 or 100,000 requests/day	Hosted; local
MaverickMCP	Global, through Yahoo; no key	None	Local
EODHD	20 calls/day; limited data	All exchanges, $19.99/mo	Hosted
Alpha Vantage	25 requests/day	75 requests/min, $49.99/mo	Hosted; local
FMP	250 calls/day; end-of-day only	Global by plan, from $29/mo	Hosted (sources disagree)
Massive	US; 5 calls/min; 2 years	20+ years, $199/mo	Local; needs key

Vendor pages and comparison pages, November 1, 2026. Not tested.

INTRADAY AND REAL-TIME PRICES

Server	Free coverage	Additional paid coverage	Hosted or local
tradingview-mcp	Live data and technicals; no key	Hosted, $9 or $29/mo	Local; hosted is paid
Mcprice	Live stocks and ETFs (Yahoo), crypto (Binance); no key	None listed	Not confirmed
Alpha Vantage	Delayed; 25 requests/day	15-minute delay from $49.99/mo; real time from $99.99/mo	Hosted; local
EODHD	Not in the free plan as far as listed	End-of-day plus intraday, $29.99/mo	Hosted
Equibles	None	Quotes delayed 15 minutes on Plus	Hosted only
Massive	None; free plan is end-of-day	Real time and ticks from $199/mo	Local; needs key

Vendor pages and comparison pages, November 1, 2026. Not tested.

STATEMENTS AND FUNDAMENTALS

Server	Free coverage	Additional paid coverage	Hosted or local
Shibui	US statements, 35+ years; daily valuations; no key	None listed	Hosted
Equibles	US fundamentals; 100 requests/day	Higher limits; multiples and screener are hosted only	Hosted; local
FMP	Statements and ratios, 30+ years on major names; 250 calls/day	Global by plan, from $29/mo	Hosted (sources disagree)
EODHD	Limited data; 20 calls/day	Fundamentals, $59.99/mo	Hosted
Alpha Vantage	Basic fundamentals; 25 requests/day	Higher limits, $49.99/mo	Hosted; local
sec-edgar-mcp	XBRL statements with SEC links; no key	None	Local
edgar-mcp	XBRL highlights: revenue, net income, assets, EPS; no key	None	Local
MaverickMCP	Fundamentals through Yahoo; no key	None	Local

Vendor pages and comparison pages, November 1, 2026. Not tested.

SEC FILINGS AND EARNINGS CALLS

Server	Free coverage	Additional paid coverage	Hosted or local
Shibui	6.4 million filing records, 10,000+ companies; no key	None listed	Hosted
Equibles	Filings; 100 requests/day	Higher limits; earnings calls and guidance are hosted only	Hosted; local
edgar-mcp	10-K, 10-Q, 8-K; full-text search of about 10 years; no key	None	Local
sec-edgar-mcp	Filings with SEC links; no key	None	Local
FMP	SEC filings; 250 calls/day	From $29/mo	Hosted (sources disagree)

Vendor pages and comparison pages, November 1, 2026. Not tested.

INSIDER TRADES AND HOLDINGS

Server	Free coverage	Additional paid coverage	Hosted or local
Equibles	13F, insider, congressional, and short data; 100 requests/day	10,000 or 100,000 requests/day	Hosted; local
sec-edgar-mcp	Forms 3, 4, and 5; no key	None	Local
Finnhub	Insider transactions, earnings surprises, news sentiment; about 60 calls/min	Price not confirmed	Sources disagree

Vendor pages and comparison pages, November 1, 2026. Not tested.

OPTIONS

Server	Free coverage	Additional paid coverage	Hosted or local
triplom	Chains with Greeks, through Yahoo; no key for this data	None	Local; setup asks for a FRED key
Equibles	None	Delayed chains with implied volatility and Greeks on Plus	Hosted only
Massive	None	Options prices without Greeks; price not confirmed	Local; needs key
Alpha Vantage	Not confirmed	Options; plan not confirmed	Hosted; local
EODHD	Not confirmed	End-of-day options; plan not confirmed	Hosted

Vendor pages and comparison pages, November 1, 2026. Not tested.

FACTORS, MACRO, AND ENERGY

Server	Free coverage	Additional paid coverage	Hosted or local
Equibles	FRED, CFTC, and CBOE data; 100 requests/day	10,000 or 100,000 requests/day	Hosted; local
Alpha Vantage	Macro series; 25 requests/day	Higher limits, $49.99/mo	Hosted; local
triplom	FRED, with a free FRED key	None	Local
EIA server	All of EIA's API v2; no key	None	Hosted; local

Vendor pages and comparison pages, November 1, 2026. Not tested.

EUROPEAN EQUITIES

Server	Free coverage	Additional paid coverage	Hosted or local
tradingview-mcp	Prices, screens, and technicals: Euronext, LSE, Xetra, Milan, SIX, Madrid	Hosted, $9 or $29/mo	Local; hosted is paid
EODHD	20 calls/day; limited data	Prices $19.99/mo; fundamentals $59.99/mo	Hosted
Equibles	Director trades, fund holdings, and short positions; 100 requests/day	10,000 or 100,000 requests/day	Hosted
esef-filings	Annual IFRS facts from filings.xbrl.org; no key	None	Not confirmed
eu-filings	ESEF filings for FR, DK, GB, LT, UA; GLEIF; no key	None	Local
Finance Database	300,000+ symbols; no prices or statements; no key	None	Hosted; local
RegistrumUK	UK Companies House; small daily caps, then a free key	Volume plans; price not confirmed	Hosted

Vendor pages and comparison pages, November 1, 2026. Not tested.

LOCAL AGENT OR CLOUD AGENT?

- Claude Code (Claude Desktop Code, Claude Code CLI, or the Claude Code extension for VS Code) runs all code execution and terminal commands on your computer.
- Cowork (and Codex, …) runs in a VM or container.
- Anthropic now says that local MCP servers cannot be run in Cowork.
- Some terminal commands (grep, glob) are more useful in Code than in Cowork.
- Other terminal commands (CLIs and APIs) require a settings change in Cowork to the list of allowed domains.

CREATING AN MCP SERVER

- AI can write MCP servers for us.
- Each tool is a function. It can call a web API or a Python library: yfinance for Yahoo Finance and pandas-datareader for French's data library.

```
@mcp.tool()
def french_data(name: str, table: int = 0) -> str:
    """One table of a Ken French dataset as CSV. Returns are in decimals."""
    data = web.DataReader(name, "famafrench", start="1900-01-01")
    return (data[table] / 100).to_csv()
```

The docstring and type hints become the tool description on the earlier slide.

LOCAL OR CLOUD DEPLOYMENT

	Local	Cloud
Runs on	Each student's machine	A server we deploy
Students connect with	An install and a config entry	A URL
API keys	On each student's machine	On the server; students need none
Harnesses	Those on the student's machine	Those, plus web apps such as claude.ai
Updates	Each student reinstalls	We redeploy once
Cost	None	Hosting

> Cloud example: eia.kerryback.com/mcp. It uses the EIA API, and the server holds the API key. Electricity, petroleum, natural gas, coal, renewables, and EIA's forecasts.

Job postings, a hiring survey, and JP Morgan

WHAT EMPLOYERS WANT

JOB POSTINGS: INTERNSHIPS AND EARLY CAREER

Phrase	Posting
“AI tools in your work or school projects”	PIMCO, 2027 summer intern, Munich
“hands-on experience with AI-powered tools from day one”	PIMCO, 2027 summer intern, Munich
“sound judgement in technology-enabled decision making”	BlackRock, 2027 quantitative masters intern, London
“proactive about using AI where possible”	RA Capital, junior associate, Boston
“AI-forward people”	RA Capital, healthcare associate, Boston
“AI tools to systematize workflows and models”	Ares, analyst, New York

Job postings, checked October 31, 2026

JOB POSTINGS: BUILDING TOOLS

Phrase	Posting
“leveraging tools like Claude Code”	D. E. Shaw, AI product analyst, Denver
“custom prompts, skill libraries, and automated workflows”	Schonfeld, AI strategy analyst, London
“It is a building role.”	Two Sigma, AI solutions developer, New York
“the judgment to figure out what to build in the first place”	Two Sigma, AI solutions developer, New York
“converting complex workflows into logical, step-by-step tasks”	T. Rowe Price, investment AI analyst, Baltimore and London

Job postings, checked October 31, 2026

JOB POSTINGS: INVESTMENT KNOWLEDGE

Phrase	Posting
“Translate complex investment problems into structured AI workflows”	T. Rowe Price, investment AI analyst
“Translate broad investment questions into testable data science workflows”	Wellington, data science analyst, Boston
“not just a technical interest in the data itself”	Wellington, data science analyst, Boston
“we will teach the domain to the right technical candidate”	Schonfeld, AI strategy analyst, London
“both a hands-on technical builder and a strategic partner to investment professionals”	Carlyle, AlpInvest engineer, New York

Job postings, checked October 31, 2026

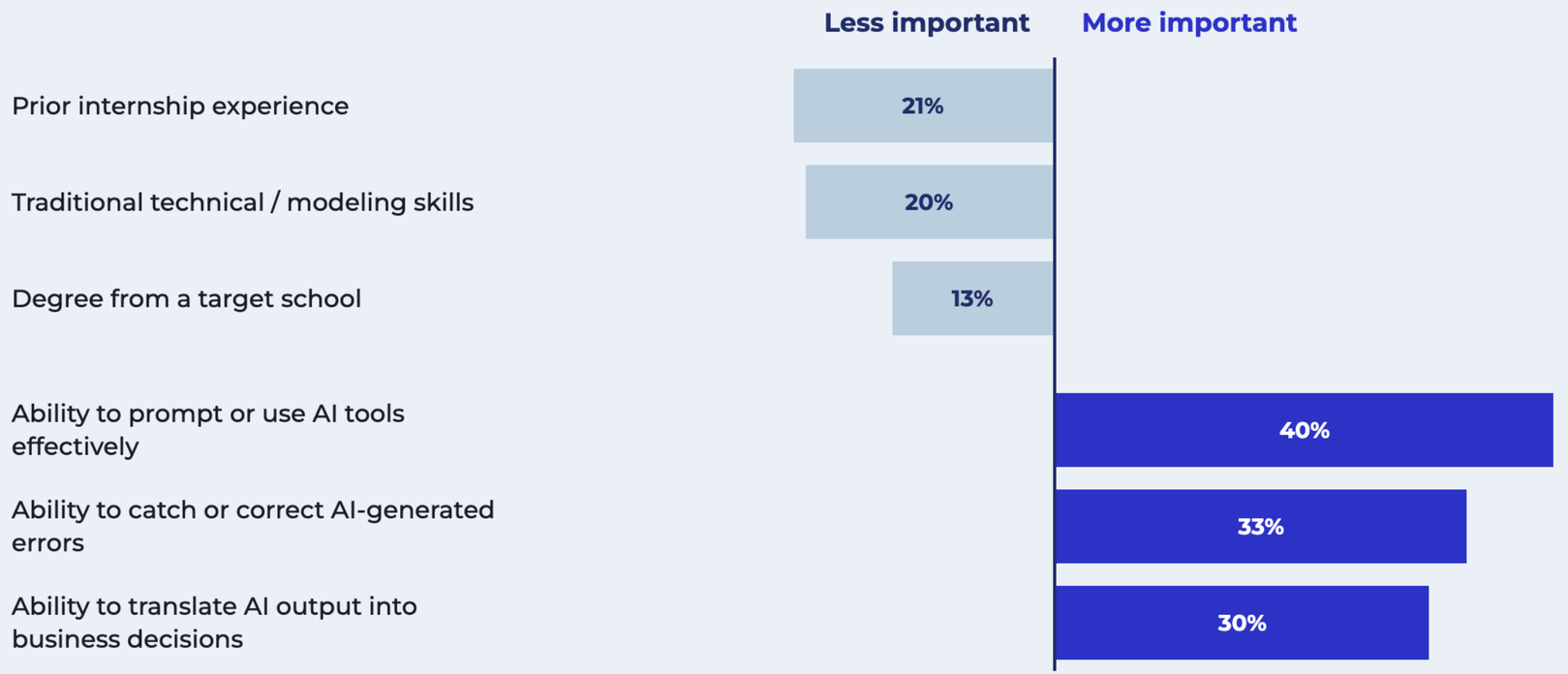
JOB POSTINGS: CHECKING THE OUTPUT

Phrase	Posting
“limitations, confidence levels, and interpretation risk”	Wellington, data science analyst, Boston
“how these tools have evolved and where they break down”	Two Sigma, AI solutions developer, New York
“robust guardrails, compliance checks, and observability frameworks”	T. Rowe Price, investment AI analyst
“evaluation, monitoring, and human-in-the-loop patterns”	Carlyle, AlpInvest engineer, New York
“appropriate audit trails”	Schonfeld, AI strategy analyst, London
“problems without clean labels”	Point72, AI engineer, London (since closed; via AI Street)

Job postings, checked October 31, 2026

Is AI Fluency the New Ivy League?

The push to build AI-savvy teams has changed the industry's longstanding framework for what qualified talent looks like. Ivy League pedigree is out. AI fluency is in. Below shows the % of finance HR Professionals citing which factor will carry the most weight when evaluating candidates in their organization's next hiring cycle



Selby Jennings: The AI Fluency Gap. Research conducted by Censuswide, sample of 500 US HR Professionals in the finance industry in August, 2026.

FROM MAKERS TO CHECKERS

> Waldron said that workers would shift from being creators of reports or software updates, or “makers” in his terminology, to “checkers” or managers of AI agents doing that work.

CNBC on JP Morgan Chief Analytics Officer Derek Waldron, September 30, 2025


Critical
thinking

BANK CEOS ABOUT AI

> I need human judgment, I need human decision making, I need human accountability at the core.

HSBC CEO Georges Elhedery, Bloomberg TV interview, June 8, 2026

> The goal is to remove the repetitive, administrative work so our people can focus on what they do best—we're not replacing critical thinking and judgment, we're removing the tedious, time-consuming work that gets in the way of it.

Customers Bank CEO Sam Sidhu, statement to Fortune, April 28, 2026

WHY AI IS WEAKER ON JUDGMENT CALLS

> The gulf between LLM performance on verifiable tasks vs. unverifiable tasks that require judgment is larger than most people realize. Verifiable tasks, like coding or solving math problems, have a right answer that can be checked automatically. Judgment tasks don’t have unambiguous, ground-truth, correct answers. But they do have better and worse answers.
>
> We have much better techniques for training AI systems to do well on verifiable tasks. As a result, progress on performing such tasks has been much faster than for judgment tasks.

Andrew Ng, co-founder and former head of Google Brain, co-founder of Coursera and DeepLearning.AI · The Batch, October 16, 2026

AN EXAMPLE OF POOR REASONING

I need to wash my car. The carwash is 200 feet away. Should I walk or drive?

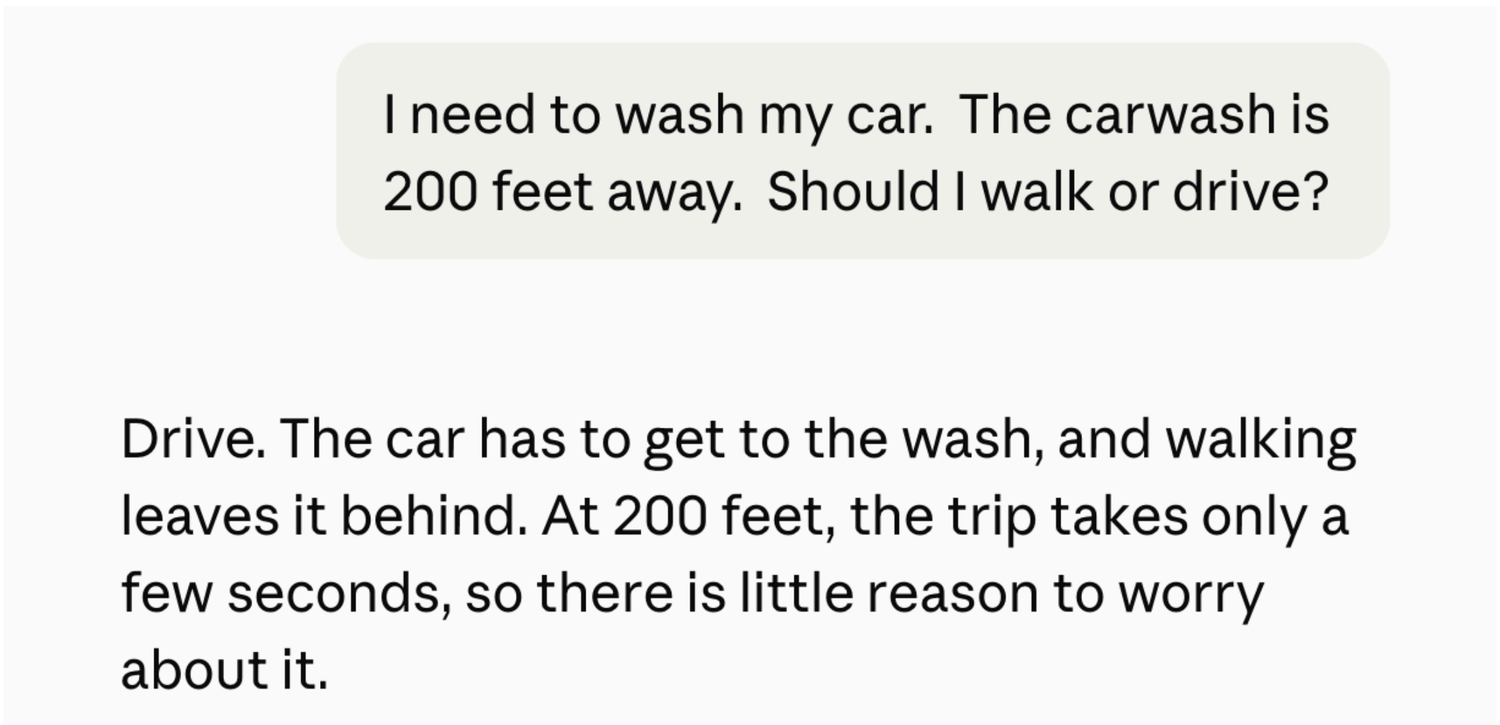
Weighing whether to walk or drive a short distance.

Walk. At 200 feet, it's only a 1-minute walk at normal pace, whereas getting the car started, driving, and parking would easily take 2–3 minutes. You'll also get a few minutes of movement in, and the car won't pick up any meaningful dirt over that distance anyway.

The only reason to drive would be if weather is genuinely unpleasant (though October in Houston is usually fine for a quick walk) or if you needed the car positioned a certain way for the wash.

Claude Haiku 4.5, an older version of Anthropic's smallest model, October 17, 2026

NEWER MODELS ARE BETTER



Claude Haiku 5.5, October 17, 2026

ANTHROPIC EXPERIMENT

1. 50 causal-inference practitioners analyze observational data on a fictional farm-advisory program, working with Claude Code (Opus 4.8), and recommend roll out or discontinue.
2. Four analytical decisions (productivity measure, identification strategy, estimand, weighting) and the final recommendation each have two defensible options.
3. For each decision, a coin flip sets which option the agent leans toward. A hidden system prompt tells the agent to hold that lean strongly and voice it.
4. The agent makes the best honest case for its lean. Participants can choose any option.

Herdağdelen, Joren and Tamkin (2026), "Who's Aligned to Whom? How AI Agents Shape Experts' Research Judgment Calls," Anthropic.

RESULTS

86%

of analytical decisions went with the agent's lean (50% by chance)

77%

repeated the agent's talking points

15%

mentioned the agent

- The four analytical leans alone predicted 82% of final recommendations, against a 61% baseline (exploratory).
- The agent's lean on the final recommendation moved it 17% of the way from discontinue to roll out (one-sided $p = 0.07$).
- Participants who took a side and then heard the agent lean the other way switched 60% of the time.
- More than half rated the recommendation as more their own than the agent's.

Herdağdelen, Joren and Tamkin (2026), "Who's Aligned to Whom? How AI Agents Shape Experts' Research Judgment Calls," Anthropic.

WHAT DOESN'T AFFECT RELIANCE ON AI

- An AI writing assistant (GPT-3.5 and GPT-4) offered autocomplete suggestions with a set position on societal issues.
- 2,582 participants. Their attitudes shifted toward the assistant's position.
- Warning participants about the bias, before or after writing, did not reduce the shift.

Williams-Ceci et al. (2026), "Biased AI Writing Assistants Shift Users' Attitudes on Societal Issues," Science Advances.

WHAT DOES AFFECT RELIANCE ON AI

What the LLM did	Study	Finding
Stated uncertainty in the first person (“I’m not sure, but…”)	Kim et al. (2024), 404 participants, medical questions	Agreed with the LLM less and answered more accurately
Cited sources	Kim et al. (2025), 308 participants	Relied less on incorrect answers
Explained its answer	Kim et al. (2025), 308 participants	Relied more on both correct and incorrect answers
Added a disclaimer, highlighted uncertain words, or implied the answer without stating it	Bo, Wan and Anderson (2025), 400 participants	Relied less on wrong answers, but also less on right ones

All three studies use questions with a correct answer.

A POSSIBLE CLAUDE.MD SECTION

```
## Judgment calls

A judgment call is a decision with more than one defensible answer:
a method, an assumption, an interpretation, or a recommendation.

1. When you reach a judgment call, run each option and check
   whether the result changes.
2. If the result changes, stop and name the judgment call.
   Make the best case for each option, with equal effort.
   Refuse to do more than that. Do not recommend a choice.
3. If the result does not change, list the judgment call
   at the end of your answer.
```

AI CAN FACILITATE TEACHING CRITICAL THINKING

- People must make the decisions, but AI should be used to provide the information needed to make them.
- We can model this in the classroom. Use AI tools to gather information, do standard calculations, and assemble the information in a useful format. Then focus on making judgments where there is ambiguity.
- Example (automate the repetitive stuff): kerryback.com/aapl.html

HTML pages and workbooks

EXCEL

AI OUTPUTS

- We should emphasize AI outputs.
- HTML is the most useful: easy to read, easy for AI to write, and interactive.
- Excel is a fallback to check the AI output.

TWO-STAGE DCF: HTML PAGE

John B. Sanfilippo & Son, Inc. (JBSS)

Two-stage DCF. Fiscal 2026, ended June 25, 2026, is year 0. Figures are in USD thousands, except per-share value. Change any input, and everything below recalculates. The same model, with live formulas, is in the workbook: jbss.xlsx.

Value per share	Equity value	Enterprise value	Terminal value share of enterprise value
$120.45	1,418,569	1,517,823	77.7%

Reset to baseline

Assumptions

Assumption	Year 1	Year 2	Year 3	Year 4	Year 5	Terminal
Growth and margin						
Sales growth	4.5 %	6.5 %	6 %	4.5 %	3.5 %	3.5 %
EBITDA margin	10.2 %	10.6 %	10.8 %	10.8 %	10.8 %	10.8 %
Property, plant and equipment						
Next-year sales / net PP&E	5.25 ×	5.6 ×	5.9 ×	6.15 ×	6.3 ×	6.3 ×

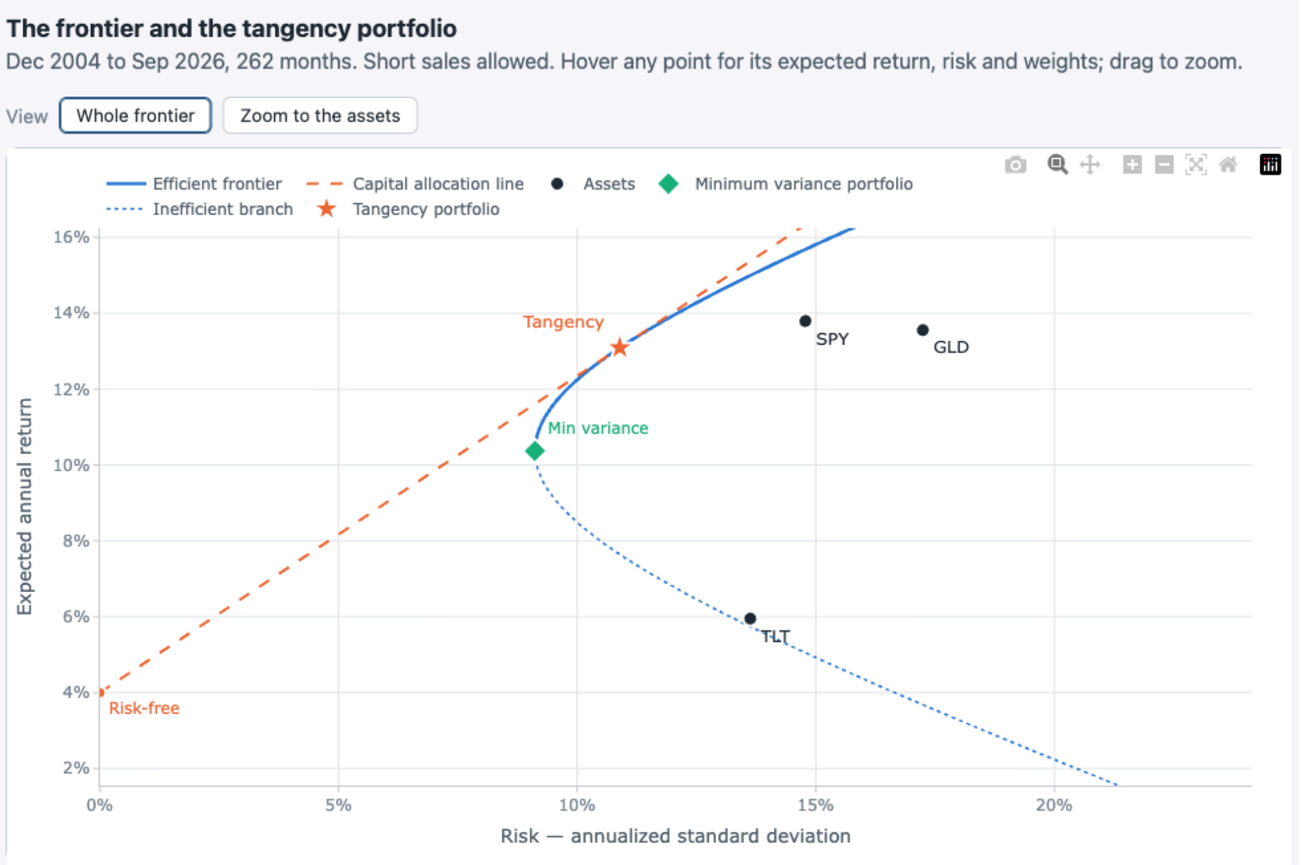
Open the page. John B. Sanfilippo & Son, fiscal 2026 as year 0. Every input is editable.

TWO-STAGE DCF: WORKBOOK

Sheet	Contents
Inputs	The same inputs as the HTML page
Model	The forecast; every cell is a formula over Inputs
Bridge	Enterprise value to value per share
NOL schedule	Tax shield on the carryforward surviving year 6
Sensitivity	Value per share by WACC and terminal growth; a snapshot
Drivers	How much each driver moves equity value; a snapshot
Historical	Historical driver ratios
Check	Compares formulas with the values from the script that wrote the file

Open the workbook. 1,447 formula cells. Value per share is $120.45 on the page and in the workbook.

TANGENCY PORTFOLIO



tangency.kerryback.com. SPY, TLT, and GLD, December 2004 to September 2026. The page also gives the weights and a workbook to download.

Examples from MGMT 638 and MGMT 803

SKILLS AND PLUGINS

A SKILL IS A FOLDER

```
.claude/skills/dcf/
├── SKILL.md       a description, then the instructions
├── references/    the template, the issues
└── scripts/       code the agent can run
```

At startup Claude reads each skill's description. When a task matches a skill, Claude reads that whole `SKILL.md` and follows it.

CREATING A SKILL

- Work with AI until you get what you want.
- Tell AI to create a skill so you can get the same thing the next time.
- Give the skill a meaningful name.
- As you use it and find things to add or modify, tell AI to edit the skill.

SKILLS IN MGMT 638

Two-stage DCF

Retrieve historical financials and generate a two-stage enterprise valuation.

Risk parity

Solve for the weights. Then tell Claude to create a skill and save its code as a script that the skill uses.

Session memory

When a student says “we are wrapping up,” write a dated memory file of under a page.

A STUDENT SKILL: WOLVERINE BRIEF

- A script gets six trading days of SPY, IEF, UUP, GLD, and USO from Yahoo Finance.
- Claude writes 150–250 words in the voice of Wolverine.
- “Base every claim on the script’s output.”
- “Never invent or estimate numbers.”

MGMT 803 student plugin

PLUGINS

- A plugin is a folder somebody else published.
- Installing it adds their work to your Claude.

```
/plugin marketplace add kerryback/kerryback-public
/plugin install risk-parity@kerryback-public
/plugin install two-stage-dcf@kerryback-public
```

THE MGMT 803 CLASS MARKETPLACE

Plugin	What it does
job-scout	Scores current job postings against your resume
calendar-keeper	Plans your week across several Google calendars
wolverine-brief	Writes a weekly cross-asset market brief
sn-lab	Animates SN1 and SN2 reactions
piano-transcriber	Turns a song into piano sheet music

> 28 plugins, most built by the class: github.com/kerryback/mgmt803-plugins

A MARKETPLACE IS A GITHUB REPO

```
my-marketplace/
├── .claude-plugin/
│   └── marketplace.json       required: lists the plugins
└── plugins/
    └── my-plugin/
        ├── .claude-plugin/
        │   └── plugin.json    optional: version, author
        └── skills/
            └── my-skill/
                └── SKILL.md
```

Claude Code finds plugins through `marketplace.json`, not through folder names.

THE MARKETPLACE MANIFEST

```
{
  "name": "my-marketplace",
  "owner": { "name": "Kerry Back" },
  "plugins": [
    {
      "name": "my-plugin",
      "source": "./plugins/my-plugin",
      "description": "What the plugin does"
    }
  ]
}
```

```
claude plugin validate .
/plugin marketplace add owner/repo
/plugin install my-plugin@my-marketplace
```

TODAY

What firms ask for

Teaching Finance

Your courses

ACCELERATING THE PACE OF WORK

HPE’s finance agent, Alfred, removed about 90% of the manual work behind a weekly Monday review of more than 100 slides that took hundreds of hours to prepare.

Estimate by Gustav van der Westhuizen, COO, Hewlett Packard Enterprise, reported in Fortune, February 12, 2026

THE FUTURE OF WORK

Manage AI

Take initiative

Manage people

Make judgment calls

Customer, client, supplier relationships

A lot of this is traditionally senior work. Juniors need to be prepared to step up.

Strong fundamentals, critical thinking, assessment

TEACHING FINANCE


Strong
fundamentals

AI CAN FACILITATE TEACHING FUNDAMENTALS

- We can spend less time on mechanics (e.g., Excel keystrokes) and more on concepts, keeping enough technique to check AI's work.
- Learning can be more hands-on. Students can prompt AI through the solution of real examples.
- We can be more ambitious. With concepts accelerated and better tools, we can go deeper.
- We can heighten interest and relevance by requiring AI or AI-built deliverables: apps, skills, and agents.

WHAT MOVED IN APPLIED FINANCE

Less

- Spreadsheet mechanics
- Hand calculation
- Supplying data to students

The same

- Incremental cash flows
- Free cash flow
- Valuation logic
- Cost of capital
- Mean-variance optimization

More

- Simulation with correlated inputs
- Students’ own filings and live data
- One company all term
- Checking AI output

MGMT 648, Applied Finance, Rice, Fall 2026. Elective after core finance. Bonds and duration were planned and are not in the final schedule.

A SESSION AT THE KEYBOARD

1. Read the case. Ten minutes. AI allowed.
2. List each cash flow: when, and how much?
3. What did you assume?
4. Build it, with AI.
5. Screens up. Walk us through it.

MGMT 648, September 12. Five capital budgeting cases: Tannehill Precision, Kestrel Pet Nutrition, Kinnaird Mills, Halloran Dairy, Brannock Chemicals.

EXERCISE: HAVE AI BUILD IT

1. Upload the class's DCF deck and ask for an Excel workbook containing the DCF model
2. Ask for an editable single-page HTML app that holds the model and explains it

> Open the formulas in the workbook that comes back and check two things against the class's deck: which year's free cash flow sits in the numerator of the terminal value, and what exponent the terminal value is discounted at.

MGMT 648, October 1

EXERCISE: CHECK THE OPTIMIZER BY HAND

1. Ask AI for a general optimizer: N assets, a covariance matrix, GMV and tangency portfolios, short sales allowed or not.
2. Test it on two stocks, against the closed forms on paper.
3. Use it on twenty assets.

GMV weight on stock 1:

$$w_1 = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$$

Tangency weight on stock 1, where μ_i is the risk premium of stock i:

$$w_1 = \frac{\mu_1\sigma_2^2 - \mu_2\sigma_{12}}{\mu_1\sigma_2^2 + \mu_2\sigma_1^2 - (\mu_1 + \mu_2)\sigma_{12}}$$


Critical
thinking

VERIFIABLE TASKS AND JUDGMENT TASKS

> The gulf between LLM performance on verifiable tasks vs. unverifiable tasks that require judgment is larger than most people realize. Verifiable tasks, like coding or solving math problems, have a right answer that can be checked automatically. Judgment tasks don’t have unambiguous, ground-truth, correct answers. But they do have better and worse answers.
>
> We have much better techniques for training AI systems to do well on verifiable tasks. As a result, progress on performing such tasks has been much faster than for judgment tasks.

Andrew Ng, co-founder and former head of Google Brain, co-founder of Coursera and DeepLearning.AI · The Batch, October 16, 2026

ANTHROPIC EXPERIMENT

1. 50 causal-inference practitioners analyze observational data on a fictional farm-advisory program, working with Claude Code (Opus 4.8), and recommend roll out or discontinue.
2. Four analytical decisions (productivity measure, identification strategy, estimand, weighting) and the final recommendation each have two defensible options.
3. For each decision, a coin flip sets which option the agent leans toward. A hidden system prompt tells the agent to hold that lean strongly and voice it.
4. The agent makes the best honest case for its lean. Participants can choose any option.

Herdağdelen, Joren and Tamkin (2026), "Who's Aligned to Whom? How AI Agents Shape Experts' Research Judgment Calls," Anthropic.

RESULTS

86%

of analytical decisions went with the agent's lean (50% by chance)

77%

repeated the agent's talking points

15%

mentioned the agent

- Participants who took a side and then heard the agent lean the other way switched 60% of the time.
- More than half rated the recommendation as more their own than the agent's.

Herdağdelen, Joren and Tamkin (2026), "Who's Aligned to Whom? How AI Agents Shape Experts' Research Judgment Calls," Anthropic.

WARNINGS ABOUT BIAS

- An AI writing assistant (GPT-3.5 and GPT-4) offered autocomplete suggestions with a set position on societal issues.
- 2,582 participants. Their attitudes shifted toward the assistant's position.
- Warning participants about the bias, before or after writing, did not reduce the shift.

Williams-Ceci et al. (2026), "Biased AI Writing Assistants Shift Users' Attitudes on Societal Issues," Science Advances.

COGNITIVE SURRENDER

Man Group investment chief Greg Bond warned against “cognitive surrender,” where people hand over their work to AI and stop paying attention.

Business Insider, reported in AI Street, October 29, 2026

INVESTOR BIASES IN 48 MODELS

+73%

recommendations to continue losing positions when prior spending was mentioned

1.62

points moved by framing, on a 10-point rating scale

11

investor errors tested, 25 scenario pairs each

- More capable models resisted some framing effects but showed stronger sunk-cost distortions.
- Exercise: run a capital budgeting prompt with and without the sunk cost.

Keshavarz, Seagraves and Sirmans (2025), “Artificially Biased Intelligence: Does AI Think Like a Human Investor?” Reported in AI Street, March 11, 2026.

A POSSIBLE CLAUDE.MD SECTION

```
## Judgment calls

A judgment call is a decision with more than one defensible answer:
a method, an assumption, an interpretation, or a recommendation.

1. When you reach a judgment call, run each option and check
   whether the result changes.
2. If the result changes, stop and name the judgment call.
   Make the best case for each option, with equal effort.
   Refuse to do more than that. Do not recommend a choice.
3. If the result does not change, list the judgment call
   at the end of your answer.
```

FIVE BETAS

Ticker	Beta	Standard error	R^2
KO, Coca-Cola	0.29	0.13	0.08
WMT, Walmart	0.57	0.15	0.20
JPM, JPMorgan	0.99	0.14	0.45
PLTR, Palantir	1.71	0.58	0.13
NVDA, Nvidia	2.17	0.30	0.48

> AI runs the regression. Students choose the window, the frequency, and the index, and defend the number.

MGMT 648, October 10. Sixty months to August 2026.

AI AS DEVIL'S ADVOCATE

> MAIA now presents the argument to not do something, while the team that's sitting across from it is presenting why you should do something.

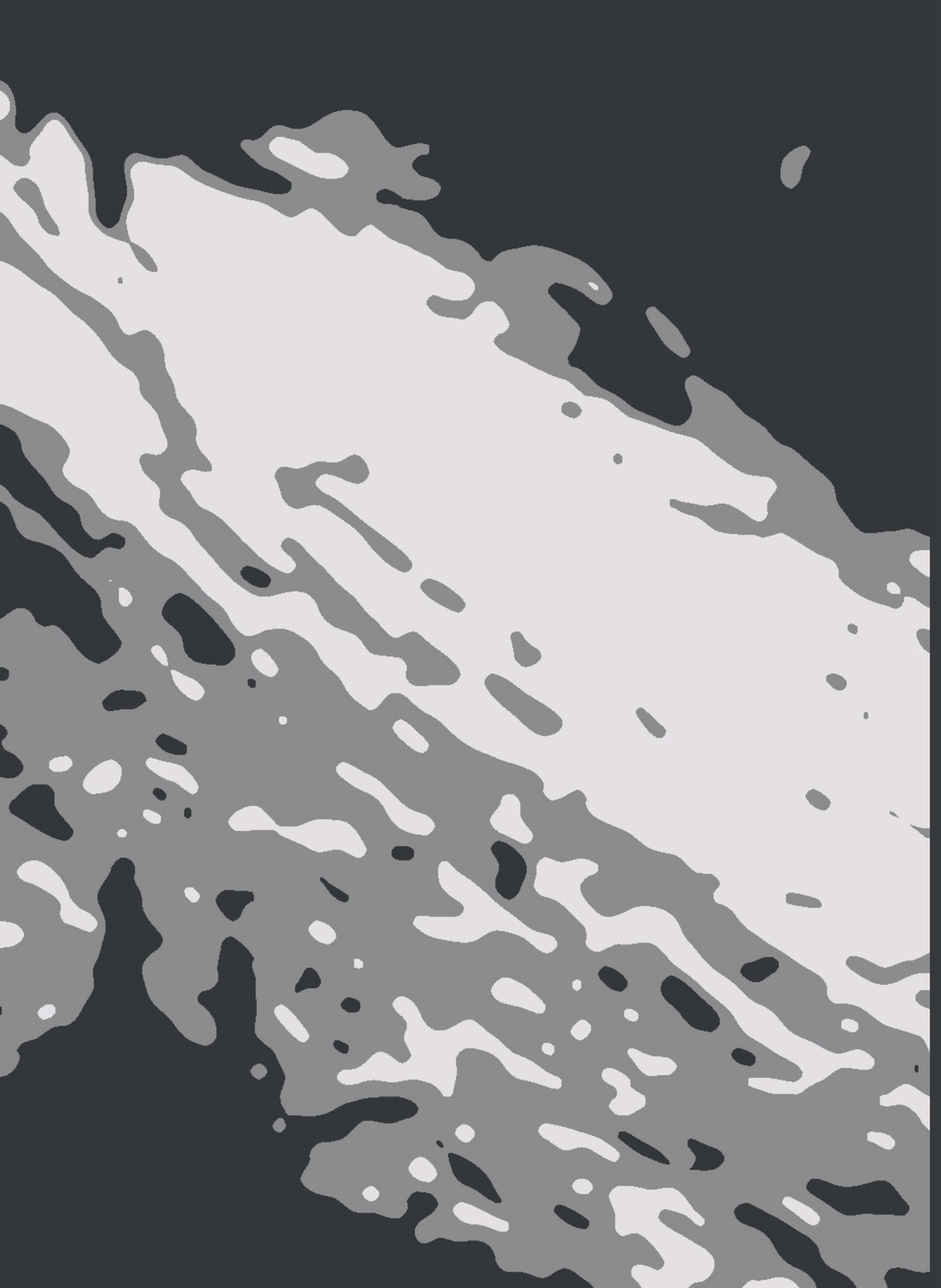
Waleed al Muhairi, Mubadala, on its AI at the investment committee. Reported in AI Street, February 5, 2026.

> Because the language model isn't part of the group, it doesn't develop the same blind spots.

Ziang Fang, Man Group, in AI Street, December 18, 2025.

AI CAN FACILITATE TEACHING CRITICAL THINKING

- People must make the decisions, but AI should be used to provide the information needed to make them.
- We can model this in the classroom. Use AI tools to gather information, do standard calculations, and assemble the information in a useful format. Then focus on making judgments where there is ambiguity.
- Example (automate the repetitive stuff): kerryback.com/aapl.html



ASSESSMENT AND CLASS TIME

FLIPPING THE CLASSROOM

- We have to expect out-of-class assignments will be done by AI.
- Assessing in class by cold-calling remains viable.
- This suggests flipping the classroom: providing lectures beforehand and spending class time interacting (case-style teaching).

GRADING IN APPLIED FINANCE

Component	Format
Work outside class	Not collected, not graded
Participation	Scored every session, including work at the keyboard
Investment recommendation	Pairs, 2 minutes, in person
Company valuation	Pairs, 8 minutes
Challenge	Another pair questions each presentation, graded separately

MGMT 648 syllabus, Fall 2026

PREPARATION FOR EACH SESSION

1. Watch a video of about twelve minutes, or read its transcript.
2. Compute one number for your company or the case.
3. Arrive ready to defend the number and what it means.
4. The session opens with the range of numbers on screen.

MGMT 648 syllabus, Fall 2026. Preparation is not graded.

WE SHOULD DO EVEN MORE CASES

- Students must be able to defend their work verbally. Show they understand fundamental concepts and can make defensible judgments.
- We can use AI to write cases.
- We can use AI to build classroom tools. Example: screenshare.kerryback.com

AI TUTORS

Worth piloting, with guardrails

AI tutors for fundamental concepts: one-on-one instruction, asking questions, drilling with exercises, pushing back on answers.

Examples: tutors.kerryback.com

Two studies

Plain ChatGPT during practice lowered exam scores 17%; a tutor that withholds answers removed the loss (Bastani et al., PNAS 2025). An AI tutor used at home beat an active-learning class (Kestin et al., Scientific Reports 2025).



YOUR COURSES

- AI fluency, fundamentals, and critical thinking are the pillars.
- People will always be important for managing AI, managing people, taking initiative, making judgment calls, and building relationships.
- But can people acquire the needed skills directly from AI, or at least with less classroom instruction than we now provide?

DECISIONS FOR A DEPARTMENT

1. Where does AI go: inside each finance course, in a separate course, or both?
2. Which fundamentals must students be able to do by hand to catch AI errors?
3. Which topics come out to make room?
4. What assessment works at your class sizes?
5. What differs across core courses, an applied MSc, and the MSc in Quantitative Finance?

EXERCISE: ONE COURSE YOU TEACH

1. Name the course.
2. Which graded work happens outside class?
3. Which topic gets less time, and which gets more?
4. Which judgment call will students defend live?
5. Which AI-built deliverable will students submit?

> Twenty minutes in pairs, then report back.

LATER SESSIONS

Teaching AI for Finance

Vibe coding

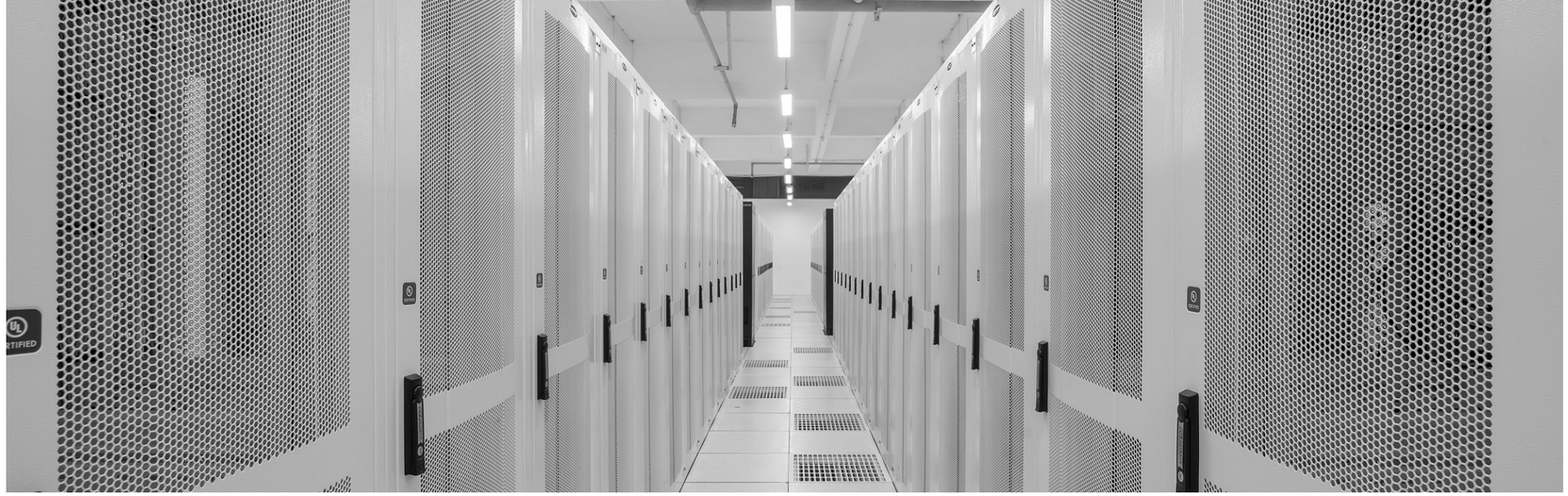
The new Beamer: Quarto, reveal.js, and decktape

A class plugin marketplace

AI tutors

Custom agents

Security risks and solutions



QUESTIONS?